Enhancing Transcription Factor Binding Site Prediction with Epigenetic Data: A BERT-Based Machine Learning Innovation

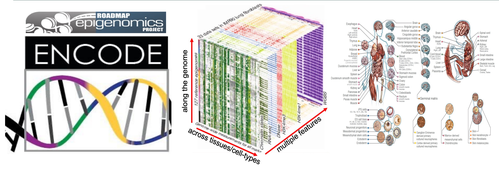
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Introduction

Background: Deciphering the functional impact of non-coding genetic variations remains a formidable challenge, especially in the context of transcription factor (TF) binding sites. These sites are crucial for understanding gene regulation but are notoriously complex to predict due to the vast and intricate nature of the human genome. Addressing this challenge, we introduce an innovative machine learning framework that integrates the BERT model, a tool originally devised for natural language processing, with advanced DNA sequence analysis. This interdisciplinary approach enables us to accurately predict TF binding sites by utilizing not only DNA sequence information but also incorporating epigenetic markers and chromatin related data. This comprehensive integration allows for a deeper understanding of the multi-layered nature of gene regulation. Particularly noteworthy is our model's proficiency in analysing non-coding regions, shedding light on their regulatory roles and impacts on TF binding, thus providing new insights into the genetic underpinnings of various diseases.

Large-Scale and Diverse Multi-Omics Datasets



Challenge: The Absence of TF ChIP-seq Data in Many Cell Types

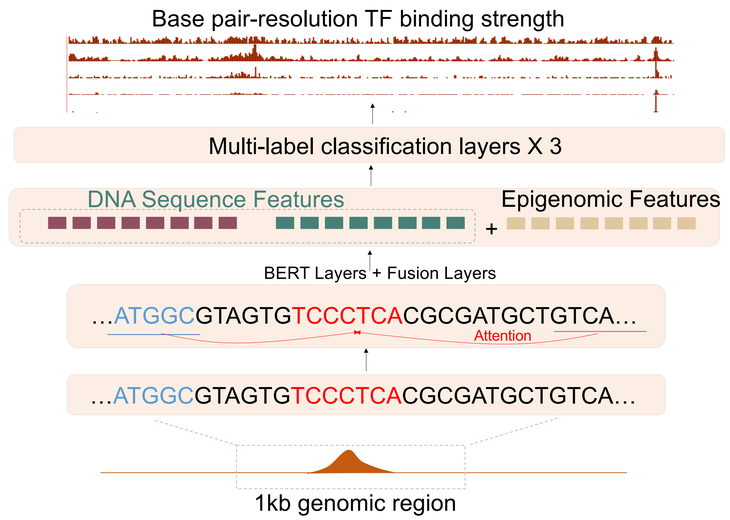
The well-characterized **K562 cell line**, including 122 TFs ChIP-seq dataset:

- **Features:** DNA Sequences & Epigenomic Histone Modifications
- **Labels:** Transcription Factor ChIP-seq Narrow Peaks

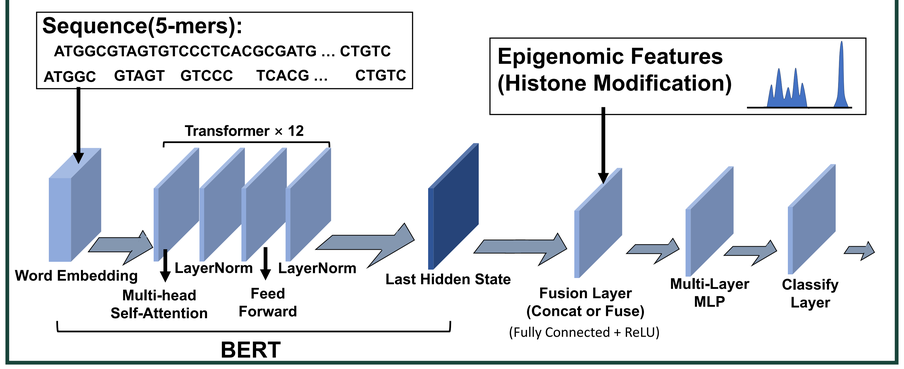
Each TF's binary classification dataset was created by adding **negative samples**, calculated as thrice the average number of binding sites, using 1kb random regions generated with BedTools.

Separation of Training and Testing Sets:
Chromosome-Based Split of Training and Testing Sets in Datasets

BERT-Based Model with Epigenetic Data



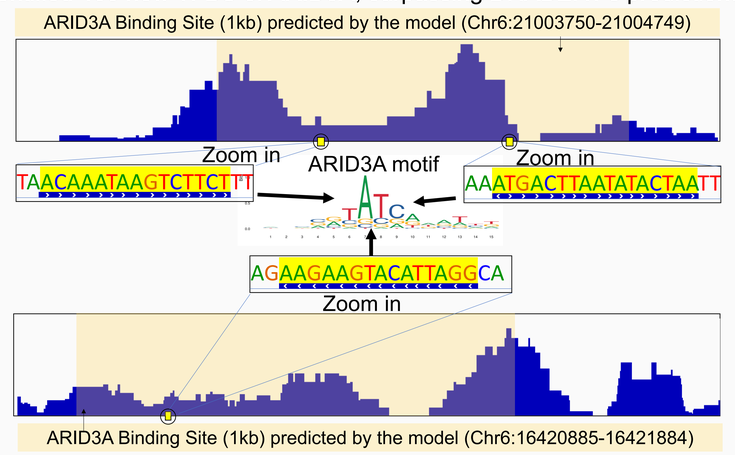
Integration of Epigenetic Features into BERT: Fuse and Concat Strategies



Newly Discovered TFBS Overlooked by DNABERT

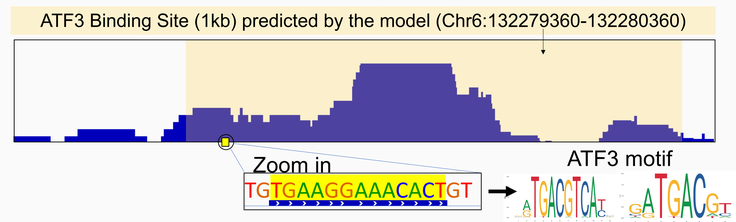
A.

Identified 460 new TFBS of ARID3A, surpassing DNABERT's predictions



B.

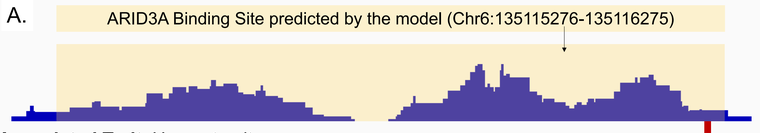
Identified 120 new TFBS of ATF3, surpassing DNABERT's predictions



Delving into the Interpretation of GWAS Variants

Newly discovered TFBS suggest potential dysregulation of gene expression, likely induced by SNPs, therefore revealing the complex mechanisms through which genetics influence cellular processes.

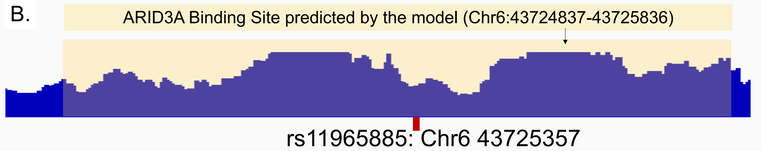
A.



Associated Trait: Haematocrit
SNP Impact: Regulatory region variant
Statistical Significance: P-value of $2*10^{-34}$, highly significant association

rs1320959: Chr6 135116248

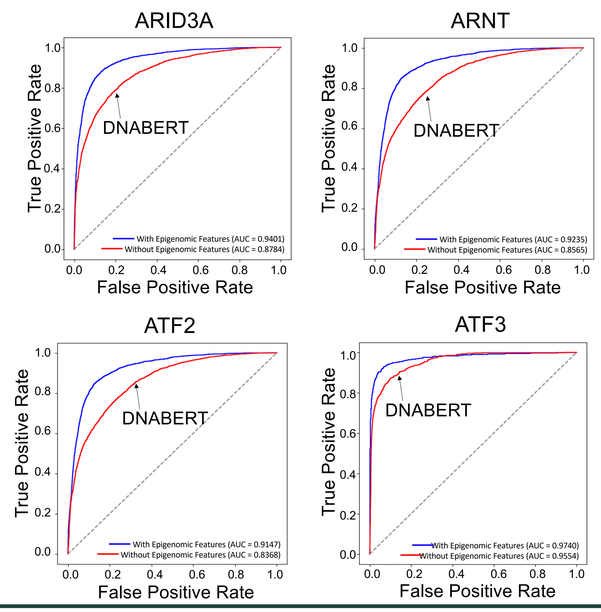
B.



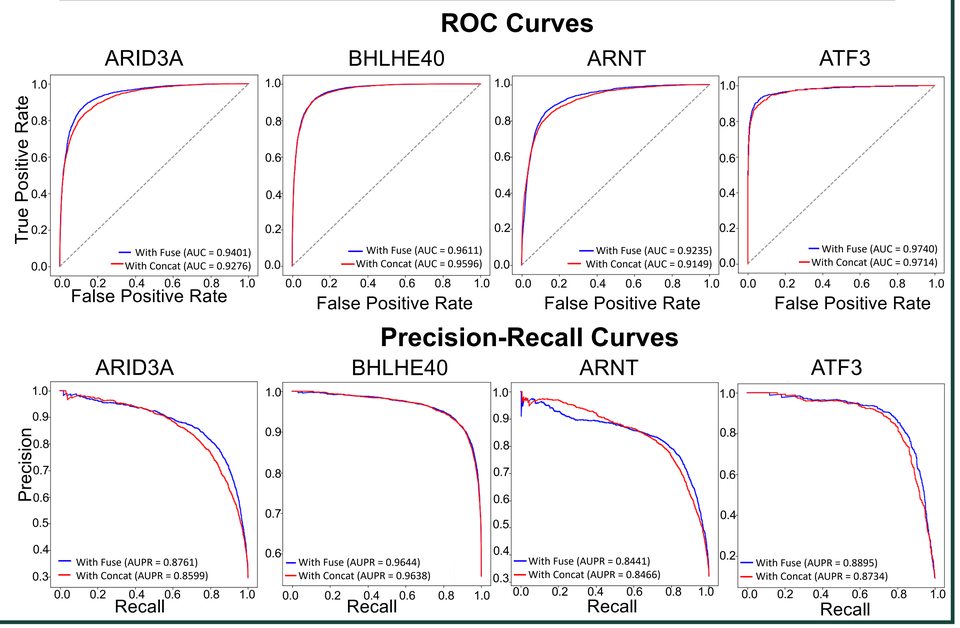
rs11965885: Chr6 43725357

Associated Trait: Reaction Time & White Blood Cell Count. **SNP Impact**: Intron variant.
Statistical Significance: P-value of $5*10^{-12}$, highly significant association

Performance Comparison with DNABERT



Performance Comparison: Fuse vs. Concat Strategies



Conclusion

TFBS Identification Enhanced: Introduces a BERT-inspired algorithm that advances the field in recognizing TFBS, overcomes ChIP-seq data scarcity, and sets innovative directions for future research.

Biological Complexity and Language Models: Adapts language models to better understand genetic intricacies, moving beyond standard applications.

Insights into Genetic Mechanisms: Connects TFBS to genetic variations, illuminating gene regulation and disease connections, suggesting paths for treatment developments.